

ORF309 Midterm 1 Q2c: Set Subtraction Explained

What Are We Looking For?

We want to find:

$$P(S_P \in [1, 2), S_N \in [1, 2))$$

This means: **Princeton gets between 1 and 2 feet (not including 2) AND New York gets between 1 and 2 feet (not including 2).**

The Notation

Define $A_{xy} = \{S_P \geq x, S_N \geq y\}$, which means “Princeton gets at least x feet AND New York gets at least y feet.”

So:

$$\begin{aligned} A_{11} &= \{S_P \geq 1, S_N \geq 1\} && \text{(both get at least 1 foot)} \\ A_{12} &= \{S_P \geq 1, S_N \geq 2\} && \text{(Princeton} \geq 1 \text{ foot, NY} \geq 2 \text{ feet)} \\ A_{21} &= \{S_P \geq 2, S_N \geq 1\} && \text{(Princeton} \geq 2 \text{ feet, NY} \geq 1 \text{ foot)} \\ A_{22} &= \{S_P \geq 2, S_N \geq 2\} && \text{(both get at least 2 feet)} \end{aligned}$$

From the problem, we know:

$$\begin{aligned} P(A_{11}) &= \frac{1}{3} \\ P(A_{12}) &= \frac{1}{4} \\ P(A_{21}) &= \frac{1}{4} \\ P(A_{22}) &= \frac{1}{5} \end{aligned}$$

The Key Insight: Set Subtraction

Let $A^* = \{S_P \in [1, 2), S_N \in [1, 2)\}$ be the event we want.

Breaking Down A^*

$S_P \in [1, 2)$ means $1 \leq S_P < 2$, which is the same as:

$$(S_P \geq 1) \text{ AND } (S_P < 2)$$

Similarly for $S_N \in [1, 2)$.

So A^* means:

- Both get **at least 1 foot** (i.e., we're in A_{11})
- **BUT** Princeton does NOT get 2 or more feet
- **AND** New York does NOT get 2 or more feet

Visualizing the Problem

Think of A_{11} as the big region where both get at least 1 foot.

Inside A_{11} , we need to **remove** cases where:

- Princeton gets ≥ 2 feet (while NY gets ≥ 1 foot) — this is A_{21}
- New York gets ≥ 2 feet (while Princeton gets ≥ 1 foot) — this is A_{12}

The Formula

$$A^* = A_{11} \setminus (A_{12} \cup A_{21})$$

In words: “Both get at least 1 foot” MINUS “at least one of them gets 2 or more feet.”

Computing $P(A^*)$

Using the set subtraction formula:

$$P(A^*) = P(A_{11}) - P(A_{12} \cup A_{21})$$

Step 1: Find $P(A_{12} \cup A_{21})$

Using the inclusion-exclusion principle:

$$P(A_{12} \cup A_{21}) = P(A_{12}) + P(A_{21}) - P(A_{12} \cap A_{21})$$

Step 2: Find $P(A_{12} \cap A_{21})$

$A_{12} \cap A_{21}$ means:

- From A_{12} : $S_P \geq 1$ and $S_N \geq 2$
- From A_{21} : $S_P \geq 2$ and $S_N \geq 1$

Taking the intersection (both must be true):

$$A_{12} \cap A_{21} = \{S_P \geq 2, S_N \geq 2\} = A_{22}$$

So:

$$P(A_{12} \cap A_{21}) = P(A_{22}) = \frac{1}{5}$$

Step 3: Plug In

$$\begin{aligned}P(A_{12} \cup A_{21}) &= P(A_{12}) + P(A_{21}) - P(A_{22}) \\&= \frac{1}{4} + \frac{1}{4} - \frac{1}{5} \\&= \frac{5}{20} + \frac{5}{20} - \frac{4}{20} \\&= \frac{6}{20} = \frac{3}{10}\end{aligned}$$

Step 4: Final Answer

$$\begin{aligned}P(A^*) &= P(A_{11}) - P(A_{12} \cup A_{21}) \\&= \frac{1}{3} - \frac{3}{10} \\&= \frac{10}{30} - \frac{9}{30} \\&= \frac{1}{30} \\&\approx 0.033\end{aligned}$$

Why This Approach Works

1. We start with A_{11} : all cases where both get at least 1 foot.
2. We subtract $A_{12} \cup A_{21}$: all cases where at least one of them gets 2 or more feet.
3. What's left is exactly the cases where **both are in $[1, 2)$** .

Common Mistake

Wrong Approach

WRONG:

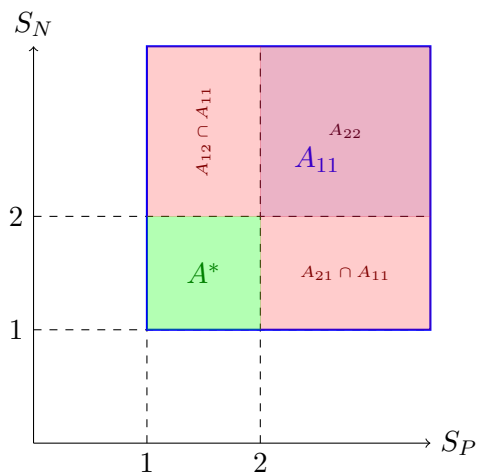
$$P(A^*) \neq P(A_{11}) - P(A_{22})$$

This would miss the cases where:

- Princeton gets ≥ 2 feet but NY is in $[1, 2)$
- NY gets ≥ 2 feet but Princeton is in $[1, 2)$

Visual Intuition

Think of it like this:



- The big blue square is A_{11} (both ≥ 1)
- The small green square is A^* (both in $[1, 2)$)
- The red regions are where at least one is ≥ 2 (we subtract these)
- The purple corner is A_{22} (both ≥ 2)

Summary

To find $P(S_P \in [1, 2), S_N \in [1, 2))$:

1. Start with $P(A_{11}) = \frac{1}{3}$ (both ≥ 1)
2. Subtract $P(A_{12} \cup A_{21})$ (at least one ≥ 2)
3. Use inclusion-exclusion: $P(A_{12} \cup A_{21}) = \frac{1}{4} + \frac{1}{4} - \frac{1}{5} = \frac{3}{10}$
4. Final answer: $\frac{1}{3} - \frac{3}{10} = \frac{1}{30}$